

Rational Functions and Asymptotes

ANSWER KEY



Finding asymptotes and intercepts

- 1 For $f(x) = \frac{x+2}{x-3}$, find:
 - a the vertical asymptote $x = 3$
 - b the horizontal asymptote $y = 1$
 - c the x -intercept $(-2, 0)$
 - d the y -intercept $(0, -\frac{2}{3})$
 - 2 Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.
Vertical: $x = 2$ and $x = -2$. Horizontal: equal degrees, so $y = \frac{2}{1} = 2$.
 - 3 Consider $f(x) = \frac{x^2-9}{x-3}$.
 - a Simplify f and identify the feature at $x = 3$. $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.
 - b Does the graph have a vertical asymptote? No — the discontinuity is removable, not an asymptote.
 - 4 Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.
Dividing: $f(x) = x + 1 + \frac{2}{x-1}$, so the slant asymptote is $y = x + 1$.
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Behaviour near asymptotes

- 5 For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:
 - a $x \rightarrow 3^+$ $f(x) \rightarrow +\infty$
 - b $x \rightarrow 3^-$ $f(x) \rightarrow -\infty$
 - c $x \rightarrow \infty$ $f(x) \rightarrow 1$
 - 6 Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.
One example: $f(x) = \frac{1}{(x-1)(x+4)}$ (any numerator of degree less than 2 works).
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Classifying end behaviour from degree

- 7 Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:
 - a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2) Degree of numerator < denominator: horizontal asymptote $y = 0$.
 - b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2) Equal degrees: horizontal asymptote $y = \frac{5}{2}$.
 - c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2) Numerator degree is exactly one more: slant asymptote (no horizontal asymptote).
- 8 Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.
Dividing: $2x^2 - 3x + 1 = (x+2)(2x-7) + 15$, so $f(x) = 2x - 7 + \frac{15}{x+2}$. Slant asymptote: $y = 2x - 7$.

Holes versus asymptotes

- 9 For $f(x) = \frac{(x-2)(x+5)}{(x-2)(x-1)}$, identify the hole and the vertical asymptote.
Common factor $(x-2)$ cancels: hole at $x = 2$ (value: $f(2) = \frac{2+5}{2-1} = 7$, so hole at $(2, 7)$). Vertical asymptote from the remaining denominator: $x = 1$.
- 10 Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.
One example: $f(x) = \frac{(x-3)(x+2)}{(x-3)(x+1)}$, which simplifies to $\frac{x+2}{x+1}$ with a hole at $x = 3$ and vertical asymptote at $x = -1$.

Reading asymptotic behavior from a graph

- 11 The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \rightarrow \pm \infty$.
As $x \rightarrow 1^-$, $f(x) \rightarrow -\infty$; as $x \rightarrow 1^+$, $f(x) \rightarrow +\infty$ — a vertical asymptote at $x = 1$. As $x \rightarrow \pm \infty$, $f(x) \rightarrow 0$ — a horizontal asymptote at $y = 0$.

Domain restrictions and sign analysis

- 12 Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.
Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Domain: $x \neq 3, -2$. On $(3, \infty)$: numerator $x-1 > 0$, denominator $(x-3)(x+2) > 0$ (both factors positive), so $f(x) > 0$ on this interval.

Asymptotes in applied contexts

- 13 The average cost per unit of producing x items is $C(x) = \frac{500+3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.
As $x \rightarrow \infty$, $C(x) \rightarrow 3$ (horizontal asymptote $y = 3$). This means as production volume grows very large, the average cost per unit approaches 3 riyals — the fixed setup cost gets spread so thin it becomes negligible.
- 14 A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2+1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.
Since the denominator has higher degree, the horizontal asymptote is $y = 0$ — the concentration eventually approaches zero as the drug is fully metabolized. $C(0) = \frac{0}{1} = 0$ mg/L (none in the bloodstream initially, before administration takes effect).

Finding a rational function's equation from its features

- 15** Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x -intercept at $x = 1$.

Numerator degree less than denominator (for $y = 0$), zero at $x = 1$: $f(x) = \frac{x-1}{(x+2)(x-3)}$.